

Plate π Recipe

A package for PLATE RECognition In PYthon

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1 Introduction

intro

2 Methods

intro

3 Examples

```
1 from platerecipy import grid, model, io
2
3 # interpolating the data onto a uniform spherical grid
4 field, coords = grid.interpolate_to_spherical(
5     xs, ys, zs,                                # Cartesian coordinates
6     edots[:, :, -1], take_log = True            # the field, changing in orders
7 )                                              # of magnitude
8
9 # creating an empty model
10 m = model.PlateModel()
11
12 # stacking the interpolated field
13 m.stack_field(field, take_log=True)
14
15 # detecting plates
16 m.find_plates(
17     interior_quantile      = 0.6,
18     wraparound_azimuthally = True,
19     min_marker_size        = 100
20 )
21
22 # storing plate DIs in various formats
23 io.save_as_vtp(m, other_fields=[coords[1], coords[2]], other_fields_names=['lat', '
24     lon'])
25 io.save_as_csv(m, other_fields=[coords[1], coords[2]], other_fields_names=['lat', '
26     lon'])
27 io.save_as_mat(m, other_fields=[coords[1], coords[2]], other_fields_names=['lat', '
28     lon'])
```